

SEMA : A Multimodal Framework for Semantic Alignment and Controlled Pictogram Generation: From JePA Semantic Representation to Stylistic Validation

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Abstract

This report introduces a novel multimodal pipeline for the automated generation of standardized pictograms, adhering to the ARASAAC visual convention. While text-to-image models have shown significant progress, maintaining strict stylistic consistency and semantic precision for assistive communication remains a challenge.

We propose a multi-stage framework that first leverages the **Pantagruel** French language model under a **Joint-Embedding Predictive Architecture (JePA)** to extract robust semantic embeddings, further refined by **Named Entity Recognition (NER)** for precise prompt structuring.

To bridge the gap between generation and validation, we developed an automated **MLP-based Judge** that evaluates image quality through a combination of edge sharpness (Brenner), visual complexity (Shannon Entropy), and semantic alignment (CLIP/MobileViT). Our results demonstrate that this architecture effectively filters non-compliant outputs, achieving a stylistic validation score of 0.88 for standardized symbols compared to 0.01 for realistic imagery.

Finally, we show that fine-tuning **Stable Diffusion (or AnyLoRA) via LoRA** on our synthetically labeled dataset enables the scalable production of high-quality, accessible visual communication tools.

Keywords— JEPA, French Language Models, NER, Multimodal Representation, Image Quality Assessment, Stable Diffusion, UNet, LoRA, Judge Models, Encoders, Self supervised learning, ARASAAC, Accessibility.

Introduction

The emergence of Large Language Models (LLMs) [Xiao and Zhu \(2025\)](#) has fundamentally redefined the landscape of Artificial Intelligence, transitioning from purely analytical tools to versatile generative engines.

While the current paradigm is dominated by autoregressive models [Xiong et al. \(2024\)](#) capable of fluid text generation and Diffusion-based architectures that have revolutionized synthetic imagery [Yang et al. \(2022\)](#), a critical challenge remains: the development of deeply grounded, context-aware encoders for specialized, high-stakes human applications

Within this generative revolution, our **SEMA** project addresses a vital societal need: the democratization of Augmentative and Alternative Communication (AAC) [Ogle-tree \(2021\)](#). For individuals with complex communication needs, translating the nuances of natural language into structured, iconographic sequences is a task that requires more than just statistical prediction ; it requires a profound semantic mapping between linguistic concepts and visual symbols.

To power this ecosystem, our framework leverages **Pantagruel** [Le et al. \(2026\)](#), a state-of-the-art family of self-supervised encoders specifically optimized for the French language [Le et al. \(2026\)](#). Rather than relying on traditional generative objectives ([Nareklshvili et al., 2024](#)) that focus on reconstructing raw inputs, Pantagruel is inspired by the **Joint-Embedding Predictive Architecture (JEPA)** [LeCun \(2022\)](#). By adopting this architecture, we benefit from a model that captures abstract semantic invariants, making it more robust for specialized downstream tasks like **Named Entity Recognition** ([Munnangi, 2024](#)).

By shifting the learning objective from token-level generation ([Nareklshvili et al., 2024](#)) to the prediction of latent representations in a shared embedding space, the JEPA framework allows the model to capture abstract semantic invariants [LeCun \(2022\)](#). This architectural choice ensures that Pantagruel develops a "**world model**" of language that is more robust to noise and better suited for downstream tasks than standard masked language models [LeCun \(2022\)](#).

The transition from a general-purpose unified model to

a specialized assistant like **Named Entity Recognition Munnangi (2024)** is achieved through a rigorous *fine-tuning* protocol Dang et al. (2025). By adapting the high-dimensional latent space of Pantagruel (Le et al., 2026) to specific linguistic benchmarks, we evaluate its ability to retain its foundational knowledge while excelling at granular classification tasks.

In this work, we propose a complete end-to-end pipeline for stylistic-consistent pictogram synthesis.

First, we address the data scarcity challenge by implementing an *automated transcription system* that generates precise textual labels from existing pictogram libraries, creating a high-quality synthetic dataset.

Second, we introduce a sophisticated **Image Validation Judge** an -Multi Layer Perceptron (MLP)-based model trained on a custom dataset of geometric and semantic metrics, to objectively evaluate stylistic compliance.

Finally, we demonstrate how this validation feedback can be integrated into the **preference fine-tuning** Dang et al. (2025) of Stable Diffusion base (SD) and AnyLoRA, via **Low-Rank Adaptation** (LoRA) Hu et al. (2021).

This approach allows for the scalable generation of standardized visual symbols that strictly adhere to the ARASAAC convention while preserving deep semantic alignment.

This article explores the convergence of these advanced architectures, demonstrating how unified predictive models (LeCun, 2022) and automated reward systems (Zhong et al., 2025) can serve as the backbone for the next generation of assistive communication technologies.

A: Pantagruel and JePA

The shift toward autonomous machine intelligence (LeCun, 2022) necessitates a move away from purely generative models (Nareklishvili et al., 2024) toward architectures that can learn robust "world models" (LeCun, 2022). Current Large Language Models (LLMs) (Xiao and Zhu, 2025) often rely on token-level reconstruction, which may under-utilize the structural regularities of continuous signals and can be disconnected from underlying reality.

1 The JePA Philosophy

The **Joint Embedding Predictive Architecture (JePA)** represents a non-generative paradigm for predictive world models (LeCun, 2022). Unlike traditional auto-encoders that attempt to reconstruct raw inputs (pixels or

words) (Hinton et al., 2006), JePA focuses on predicting the **latent representation** of unobserved parts of an input from observed parts.

As illustrated in the JEPA framework (Figure 1),

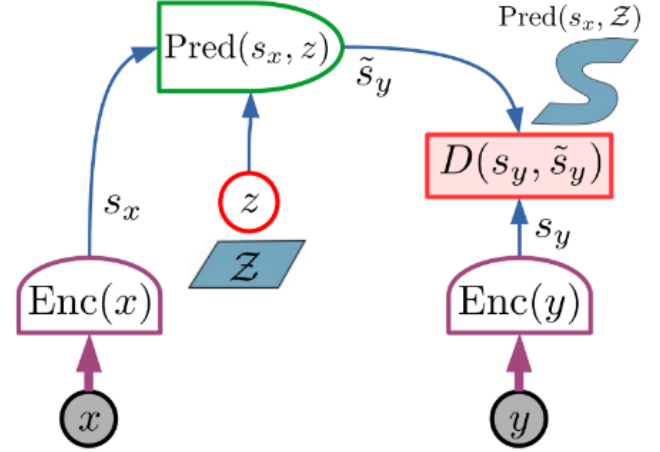


Figure 1: The Joint-Embedding Predictive Architecture (JEPA) framework. It extracts representations s_x and s_y from inputs x and y via two encoders (Enc1 not need to be same as Enc2). A predictor module estimates s_y from s_x using a latent variable z , minimizing the prediction error $D(s_y, \tilde{s}_y)$ in the feature space. Adapted from (LeCun, 2022)

the architecture extracts abstract representations that filter out unpredictable or irrelevant noise - such as background textures - to concentrate on high-level semantic features LeCun (2022). This allows the system to capture the mutual dependencies between inputs without the computational burden of generating every detail.

JEPA is fundamentally **non-generative** in the sense that it does not reconstruct the raw input y from x . Instead, it captures the dependencies between x and y by performing predictions directly in the representation space. By shifting the learning objective to the prediction of latent representations, the framework avoids the complexities of pixel-level generation Nareklishvili et al. (2024) while effectively capturing abstract semantic invariants.

Furthermore, the architecture is strictly **non-contrastive** Kuhn and Buettner (2026). Unlike contrastive learning methods Garrido et al. (2022) that require negative samples to prevent representation collapse, JEPA relies on the architectural constraint of predicting in latent space to learn meaningful features.

By shifting the learning objective to the prediction of latent representations, the framework avoids both the complexities of pixel-level generation (Nareklishvili et al., 2024) and the computational overhead of maintaining large batches of negative pairs.

2 Pantagruel: A Unified French Encoder

Pantagruel (Le et al., 2026) is a state-of-the-art family of self-supervised encoders designed specifically for the French language, utilizing the JEPA-based *data2vec 2.0* framework Baevski et al. (2023), which is a self-supervised algorithm that unifies modalities (vision, speech, and text) by predicting contextualized latent representations

As noted in the research of Meta’s team Baevski et al. (2023), data2vec 2.0 is significantly more efficient than its predecessor. While it achieves accuracy levels comparable to existing self-supervised algorithms in computer vision, it does so 16x faster. This efficiency stems from its ability to (figure 2):

- **Learn in the feature space:** Instead of reconstructing raw pixels or tokens, it predicts abstract representations, reducing the computational overhead.
- **Utilize a Teacher-Student paradigm:** The "teacher" encoder observes the full input to produce a contextualized target, while the "student" must predict this target from a masked version
- **Accelerated Training:** By leveraging the JEPA philosophy, it avoids the heavy cost of generative decoding, focusing only on the prediction of latent features

α . Pantagruel Architecture and Training

The architecture of Pantagruel (figure 3) aims to unify the treatment of text and speech by learning contextualized target representations in the feature space rather than predicting modality-tailored targets like textual tokens or speech units.

For the text modality, Pantagruel introduces a hybrid objective:

- **Latent Space Prediction (L_2 Loss):** This loss measures the L_2 distance between student predictions and the teacher encoder’s representations over masked regions. It allows the model to capture global linguistic regularities in the feature space.
- **Masked Language Modeling (MLM):** The student decoder also predicts the actual masked text tokens. This augments the objective to capture fine-grained syntactic and semantic information, which is often lost in purely latent-space models.

β . Optimization and Stability

The decoder, used only during pre-training, predicts the teacher’s latent representations from the student’s encoder

outputs.

Both encoders Hinton et al. (2006) share a standard Transformer architecture Vaswani et al. (2017). To ensure stability, the teacher’s parameters are maintained as an **Exponential Moving Average (EMA)** of the student’s:

$$\theta_{teacher} = (1 - \alpha)\theta_{student} + \alpha\theta_{teacher} \quad (1)$$

With α typically set to 0.999, the teacher remains more stable than the student, providing a consistent target for the predictive modeling paradigm LeCun (2022).

γ . Model Configurations

Pantagruel is proposed in two main configurations Vaswani et al. (2017), (figure 4):

- **Base:** 12 layers, 8 attention heads, and 768 hidden dimensions.
- **Large:** 24 layers, 16 attention heads, and 1024 hidden dimensions.

δ . Massive Data Scale

To build a robust foundation Huyen (2025), Pantagruel was pre-trained on an unprecedented scale for the French language Le et al. (2026). This massive dataset allows the model to capture a wide array of linguistic and acoustic nuances:

- **Textual Data:** The model was trained on over 30 billion tokens. The corpus includes high-quality curated data from Wikipedia (4GB), the massive OS-CAR crawl (138GB), and the bilingual CroissantLLM dataset (300GB) Faysse et al. (2024).
- **Speech Data:** A total of 114,000 audio hours were utilized. This includes the newly introduced **INA-100k corpus** from the *Institut National de l’Audiovisuel*. The INA dataset covers 80 years of diverse French radio and TV broadcasts. This significantly enhances the model’s robustness to acoustic variability, such as background noise, varying recording qualities, and evolution in speaking styles over decades.

ϵ . Implementation details

Le et al. (2026) follows the training recipe established by data2vec 2.0. The optimization process utilizes the **Adam optimizer** Diederik et al. (2014) combined with a cosine learning rate scheduler Loshchilov and Hutter (2017). As

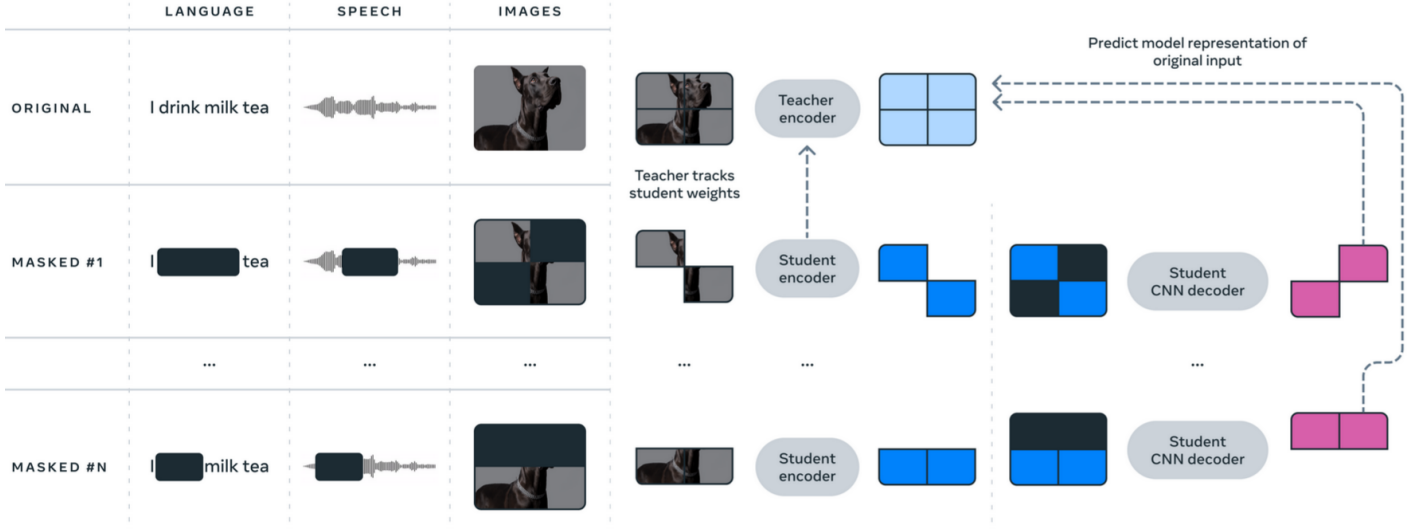


Figure 2: Mechanism of the data2vec 2.0 framework: The student encoder predicts the teacher’s representations for masked regions, stabilized by an Exponential Moving Average (EMA).

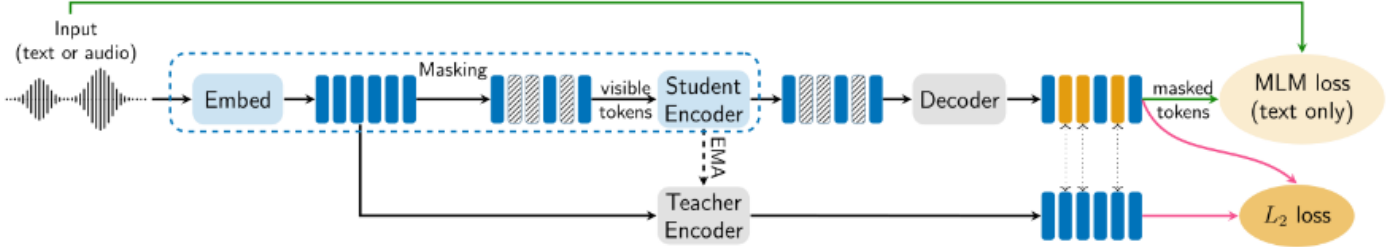


Figure 3: Overview of the Pantagruel model architecture, based on data2vec 2.0 and JePA frameworks. A modality-specific pre-Net extracts features for the teacher and student encoders. For text, an additional Masked Language Modeling (MLM) loss is combined with the L_2 representation loss [Le et al. \(2026\)](#).

illustrated in Figure 5, the scheduler operates in two distinct phases:

- **Linear Warmup Phase:** The learning rate η is increased linearly from a minimum value η_{min} to a configured maximum η_{max} over N initial steps.
- **Cosine Decay Phase:** After the warmup, η decreases smoothly following a cosine curve until the training reaches the maximum number of updates N_{max} , at which point η returns to η_{min} .

Following the implementation of [Loshchilov and Hutter \(2017\)](#) cited in the training recipe, the learning rate decay η_t is governed by the following equation during the **cosine Decay phase**:

for $t > N$:

$$\eta_t = \eta_{min} + \frac{1}{2}(\eta_{max} - \eta_{min}) \left(1 + \cos \left(\frac{t - N}{N_{max} - N} \pi \right) \right) \quad (2)$$

This specific scheduling is critical for the stability of the EMA (**Exponential Moving Average**) teacher-student update mechanism used in Pantagruel.

ζ. Benchmarks and Performance

Pantagruel demonstrates competitive or superior performance across standard benchmarks like **FLUE** (text) and **LeBenchmark** (speech). Notably, it excels in semantic tasks such as **Extractive Question Answering** (PIAF) and achieves high robustness in **Speech Emotion Recognition** (SER), [Le et al. \(2026\)](#).

These results confirm that feature-space self-supervision scales efficiently, providing the necessary semantic "intelligence" to guide the pictogram generation pipeline .

η. Named Entity Recognition Benchmark

We investigate Pantagruel’s performance on the **Named Entity Recognition** (NER) ([Munnangi, 2024](#)) benchmark and compare it with established baselines, including **CamemBERT** [Martin et al. \(2019\)](#) and **RoBERTa**

	Model name	Param	Trained on
Text	Pantagruel-B-camtok-Wk	110M	Wikipedia
	Pantagruel-B-Wk	125M	Wikipedia
	Pantagruel-B-Wk-MLM	125M	Wikipedia
	Pantagruel-B-camtok-Osc	125M	OSCAR
	Pantagruel-B-Crs-MLM	125M	croissantLLM
Speech	Pantagruel-B-1k	93M	LibriSpeech
	Pantagruel-B-14k	93M	LeBenchmark
	Pantagruel-L-14k	313M	LeBenchmark
	Pantagruel-L-114k	313M	LeBenchmark + INA

Figure 4: Pantagruel model names and configurations. The suffix “B” and “L” mean base and large, respectively. Models trained with Camembert tokenizer are denoted with “camtok”. For text models, the last suffix denotes the text dataset names, while for speech it is the size of the datasets in terms of audio hours (e.g., “1k” means 1000 hours of audio) . Adapted from [Le et al. \(2026\)](#)

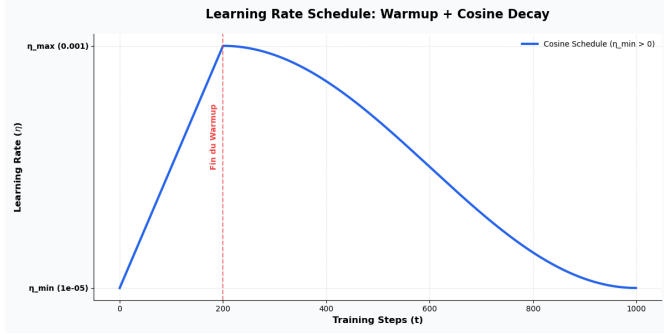


Figure 5: Cosine learning rate scheduler

[Liu et al. \(2019a\)](#). For our experiments, we use the **RoBERTa_base** model released by Facebook [Liu et al. \(2019b\)](#).

NER is a sequence labeling task where the model categorizes spans of text into predefined semantic classes

In this study, NER serves as a primary *semantic probing* task. Our objective is to evaluate the quality of the contextual representations. Specifically, we examine whether the unified pre-training of **Pantagruel** provides a superior structural understanding of French syntax compared to traditional encoders.

1. Data and Pre-processing

The models are fine-tuned on the French version of the **WikiNER** dataset, originally introduced by [Nothman et al. \(2012\)](#). This corpus, sourced from Wikipedia, is specifically designed to evaluate multilingual NER capabilities. We utilize the specific version hosted on the Hugging Face hub [Naguib \(2024\)](#), which presents a significant *class imbalance* among its four categories:

- **’O’**: Outside of an entity (appears approximately 2.6 million times).

- **’LOC’**: Locations (cities, countries, geographic features) (137 000 times).
- **’PER’**: Persons (first names, last names, or full names), (120 000 times)
- **’ORG’**: Organizations (companies, agencies, institutions), (10 000 times)

To ensure consistency across different tokenization strategies (WordPiece vs. SentencePiece), **we employ a *first-encoding alignment protocol***. For each multi-token word, the label is assigned exclusively to the first sub-token, while subsequent sub-tokens are assigned a masking index of -100 to be ignored during loss computation.

2. Models Under Evaluation

We conduct a comparative analysis of three distinct transformer-based architectures:

- **CamemBERT base**: A monolingual French model based on the RoBERTa architecture [Martin et al. \(2019\)](#).
- **RoBERTa base**: A multilingual model representing state-of-the-art cross-lingual transfer [Liu et al. \(2019a\)](#).
- **Pantagruel** : A unified model leveraging JEPA-based pre-training to enhance French text representation [Le et al. \(2026\)](#).

3. Architecture and Training

For each encoder ([Hinton et al., 2006](#)), we append a **task-specific classification head** designed to map the contextual embeddings to the **label space**. This head consists of a **Multi-Layer Perceptron (MLP)** comprising three linear layers. Each intermediate layer is followed by a **ReLU activation** function and a **Dropout layer** to mitigate overfitting.

The probability distribution over the set of labels $\mathcal{Y}$ is computed via a linear transformation followed by a Softmax activation:

$$P(y|h_i) = \text{Softmax}(Wh_i + b) \quad (3)$$

Where $W \in \mathbb{R}^{|\mathcal{Y}| \times d}$ is the weight matrix and $b \in \mathbb{R}^{|\mathcal{Y}|}$ is the bias vector. In our case $|\mathcal{Y}| = 4$.

A critical implementation detail concerns the alignment between the tokenizer’s output and the word-level annotations. Since Transformers often split a single word into multiple sub-tokens, we adopt a **first-token prediction strategy**:

1. Only the embedding of the first sub-token of each word is fed into the classification head to predict the entity label.
2. All subsequent sub-tokens (the "tails" of the words) are assigned a label index of -100 .

This ensures that the **Weighted Cross-Entropy loss** 2 ignores these indices during backpropagation, preventing the model from being penalized for sub-word fragments.

Fine-tuning is performed using the Hugging Face **Trainer** API with the following hyperparameters:

- **Optimizer:** AdamW with a weight decay of 0.01 to ensure regularization.
- **Learning Rate:** A maximum learning rate of $\eta = 2 \times 10^{-5}$ coupled with a linear decay scheduler.
- **Precision:** 16-bit Mixed Precision (FP16), enabling higher batch sizes and faster computational throughput by optimizing VRAM utilization.

4. Loss Function and Class Weighting

The training objective is to minimize the Weighted Cross-Entropy (WCE) loss Elharrouss et al. (2025), which measures the discrepancy between the predicted probability distribution and the ground-truth labels while accounting for the extreme class imbalance.

In the WikiNER dataset Naguib (2024), the background class 'O' represents approximately 0.95 of the tokens. To counteract this, we assign a weight vector W where the weight w_c for each class c is inversely proportional to its frequency. For a sequence of T tokens, the loss is defined as:

$$\mathcal{L} = -\frac{1}{N_{\text{valid}}} \sum_{i: y_i \neq -100} w_{y_i} \log(\hat{y}_{i, y_i}) \quad (4)$$

Where:

- $C = 4$ is the total number of classes (4 semantic entities plus the background class).
- w_{y_i} is the weight assigned to the ground-truth class of token i , calculated as the inverse frequency of that class in the training set ($w_c = \frac{1}{\text{freq}_c}$).
- $\hat{y}_{i, y_i}$ is the predicted probability for the correct class y_i after the Softmax layer.
- N_{valid} is the number of tokens where $y_i \neq -100$, ensuring the loss is only computed on the first sub-token of each word.

5. Evaluation Metrics

To evaluate the semantic performance of the models, we employ token-level classification metrics. Our approach focuses on the direct categorization of each word's primary sub-token into its respective semantic class or the background class 'O'.

- **Accuracy (A):** The proportion of correct predictions.
- **Precision (P):** The proportion of predicted entities that match the ground-truth class labels.
- **Recall (R):** The proportion of actual ground-truth semantic labels correctly identified by the model.
- **F1-Score:** The harmonic mean of Precision and Recall, used to mitigate the impact of the significant class imbalance in the WikiNER dataset.

$$A = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN} \quad (6)$$

$$F_1\text{-score} = 2 \cdot \frac{P \cdot R}{P + R} \quad (7)$$

- **True Positives (TP) :** The number of entity tokens (PER, LOC, ORG) correctly classified by the model.
- **False Positives (FP) :** The number of tokens predicted as an entity that are actually background text ('O'), or entities predicted as the wrong category.
- **False Negatives (FN) :** The number of actual entity tokens that the model failed to detect, classifying them as background text ('O').
- **True Negatives (TN) :** The number of non-entity tokens ('O') correctly classified as background text.

6. Results and Discussion

The comparative performance of the encoders is summarized in Table 1.

It is important to note that Pantagruel-Camtok was trained without loss weighting, whereas the other models (Pantagruel-base, RoBERTa-base, and CamemBERT-base) utilized class-frequency weights to mitigate the dataset's inherent bias.

Due to the extreme class imbalance in WikiNER (2.6×10^6 'O' instances vs. 137,000 entities), a model could easily achieve high Recall and high accuracy by over-predicting the dominant class. Therefore, **Precision** becomes the most rigorous measure (with the **F1-score**) of the model's ability to extract meaningful information without generating excessive noise.

Table 1: NER Performance Comparison on WikiNER

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Pantagruel-Camtok	96.8	69.7	80.3	74.6
Pantagruel-base	90.7	39.6	63.7	48.9
RoBERTa-base	97.4	76.5	86.6	81.2
CamemBERT-base	98.6	87.3	91.1	89.1

- **Tokenizer Impact:** Pantagruel-Camtok significantly outperforms the base version (74.6% vs. 48.9% F1-score), proving that French-specific segmentation is crucial for delineating semantic boundaries.
- **Robustness Without Weighting:** Despite being the only model trained without loss weighting, Pantagruel-Camtok maintains a solid 69.7% Precision. This demonstrates that the tokenizer’s representations are robust enough to mitigate majority-class bias without mathematical assistance.
- **CamemBERT Efficiency:** CamemBERT-base achieves state-of-the-art results with an 89.1% F1-score in only 7 epochs, underscoring the superior efficiency of its pre-trained representations.

To further analyze the training dynamics of our contribution, Figure 6 illustrates the evolution of all four metrics for the Pantagruel-Camtok model. The steady increase in Precision (69.7%), achieved without loss weighting, confirms the structural effectiveness of the Camtok tokenizer. Moreover, the visible gap between the high, stable Accuracy (96.8%) and the evolving F1-score justifies our reliance on the latter as the primary indicator of performance

[62440/62440 1:35:13, Epoch 10/10]

Epoch	Training Loss	Validation Loss	Precision	Recall	F1	Accuracy
1	0.231532	0.198895	0.482197	0.514135	0.497654	0.941586
2	0.192424	0.163829	0.517978	0.668972	0.583871	0.949857
3	0.160545	0.140828	0.592484	0.700088	0.641807	0.955949
4	0.138111	0.122789	0.625241	0.746254	0.680409	0.961038
5	0.121167	0.113756	0.648743	0.771270	0.704720	0.963925
6	0.110128	0.107090	0.671351	0.783676	0.723178	0.966101
7	0.101639	0.107414	0.673687	0.789370	0.726955	0.965984
8	0.093816	0.101347	0.699929	0.797437	0.745508	0.968333
9	0.088282	0.101984	0.700670	0.801369	0.747644	0.968977
10	0.082829	0.102311	0.696790	0.803403	0.746308	0.968689

Figure 6: Pantagruel_camtok’s metrics

The experimental results on the WikiNER dataset highlight the critical role of language-specific tokenization.

As illustrated in Figure 7, our analysis yields two primary conclusions:

- **Efficiency under Constraints:** CamemBERT-base establishes the state-of-the-art performance

(89.1% F1-score) by epoch 7. Although training was terminated prematurely due to VRAM saturation, the trajectory in Figure 7 confirms its rapid convergence and the superior efficiency of its pre-trained French representations.

- **Robustness of Camtok:** Pantagruel-Camtok shows a stable and consistent progression, reaching an F1-score of 74.6% by epoch 10. The fact that it significantly outperforms the base version (48.9%) while being the only model trained **without loss weighting** validates the superior semantic alignment provided by the Camtok tokenizer.

Ultimately, the trends observed in Figure 7 demonstrate that the performance gains in the Pantagruel architecture are driven by structural tokenization quality rather than loss-function optimization.

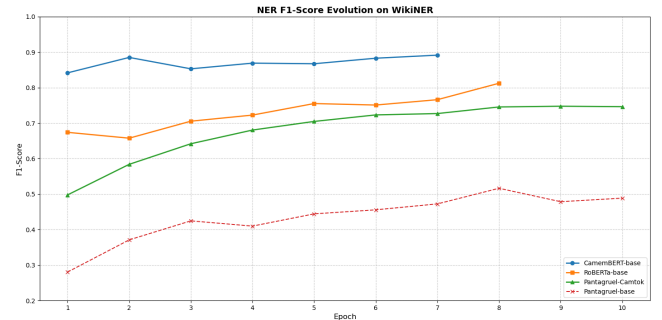


Figure 7: Evolution of F1-score across training epochs for all evaluated encoders

B : Picto and Diffusion

1 Augmentative and Alternative Communication (AAC)

Augmentative and Alternative Communication (AAC) encompasses tools and strategies to facilitate communication when the conventional language abilities of an individual are impaired or absent [Ogletree \(2021\)](#).

Language impairment affecting (oral production, comprehension, listening, reading, and writing) can arise from various sources, including genetic diseases, autism spectrum disorders, or intellectual disability.

AAC offers a range of tools, such as communication boards, signs, and computer devices, to help individuals effectively convey their messages in everyday situations.

Specifically, AAC involves the mapping of natural language [Macaire et al. \(2024\)](#) into **pictograms**: graphic symbols associated with specific concepts such as objects, persons, or actions. The use of these symbols has demonstrated its effectiveness in visualizing syntax, manipulating linguistic units, and enhancing overall language accessibility.

The ARASAAC Standard and the Semantic Unit

A pictogram is not merely a drawing but a **semantic unit**. In this framework, we adopt the **ARASAAC** visual convention as our target standard for three primary reasons (Table 2):

- **Global Standardization:** It provides a consistent and universally recognizable visual language for users with complex communication needs.
- **Mathematical Constraints:** The ARASAAC style—characterized by bold black outlines and uniform color fields—is ideal for objective quality assessment.
- **Addressing the Semantic Gap:** While robust, the official ARASAAC library is finite. Our goal is to leverage generative AI to expand this vocabulary without breaking stylistic homogeneity.

As illustrated in Figure 8, while communication boards provide a vital interface, their fixed nature limits personalized expression. Existing databases often fail to cover the full linguistic diversity of the French language, creating a "semantic gap" for users.

Table 2: Visual Consistency across ARASAAC Concepts. Each row displays four variations of a single semantic concept .

Concept	ARASAAC Pictograms			
Eat				
Happy				
pants				
House				

This limitation necessitates an automated pipeline capable of expanding the library while strictly adhering to these visual constraints

2 Related Work: Evolution of Generative Models and LORA

The field of synthetic image generation has evolved through several major shifts, moving from rigid latent space models to the high-fidelity results seen in modern diffusion systems.

α . Evolution of Generative Models

Early approaches were dominated by **Generative Adversarial Networks (GANs)** and **Variational Autoencoders (VAEs)**.

- GANs ([J.Goodfellow et al., 2014](#)) are capable of producing sharp images by training a generator against a discriminator, but they often suffer from training instability.
- VAEs ([Kingma and Welling, 2013](#)) on the other hand, provide a more stable latent space but tend to gen-



Figure 8: AAC principles: The transition from impaired oral speech to the use of iconographic symbols on a communication board to enable social interaction.

erate blurry outputs due to the limitations of their reconstruction process.

At the heart of many of these architectures is the **U-Net** [Ronneberger et al. \(2015\)](#). Originally designed for medical imaging, its symmetric encoder-decoder structure with **skip connections** ([Xu et al., 2024](#)) allows it to preserve fine details while capturing the global context of the image. This makes it an ideal backbone for denoising processes.

β . Diffusion Models and Stable Diffusion

Diffusion Models ([Yang et al., 2022](#)) represent a fundamental change by learning to reverse a gradual Gaussian noise process. The model (?) is trained to predict the noise ϵ added to an image at a specific timestep t . To reduce the heavy computational cost of working in pixel space, **Stable Diffusion** ([Oorloff et al., 2025](#)) operates within a compressed **Latent Space** $\mathcal{Z}$.

As illustrated in Figure 9, this architecture relies on three main pillars:

- **VAE**: An encoder $\mathcal{E}$ compresses the image into latent data z , and a decoder $\mathcal{D}$ transforms the denoised data back into a real image.
- **Conditioning Mechanism** (τ_θ): This block injects external information, such as text, into the U-Net through **cross-attention** layers.

But, The central engine of the diffusion process is the **Denoising U-Net** (ϵ_θ), as illustrated in Figure 9. Unlike standard feed-forward networks, this architecture is specifically designed to predict the noise ϵ added to the latent representation at each timestep t .

- **Hierarchical Structure**: The U-Net ([Ronneberger et al., 2015](#)) employs a symmetric contracting and

expanding path. The encoder extracts high-level semantic features, while the decoder reconstructs the spatial information.

- **Skip Connections** ([Xu et al., 2024](#)) : Represented by the dashed arrows in Figure 9, these connections transfer fine-grained spatial details directly from the encoder to the decoder. In our context, this is essential for maintaining the sharp, high-contrast outlines characteristic of ARASAAC pictograms.
- **Cross-Attention Mechanism** ([Vaswani et al., 2017](#)) : As seen in the Q, K, V blocks within the U-Net, the model integrates external conditioning τ_θ . By using Queries from the visual features and Keys/-Values from the semantic labels, the U-Net effectively "grounds" the generated pixels into the intended concept.

In this setup, **Queries** (Q) represent visual features in the U-Net, while **Keys** (K) and **Values** (V) represent the semantic information from the text. This allows the model to "guide" the creation of pixels to match the intended concept. The "switch" shown in Figure 9 highlights the model's flexibility to receive different types of guidance.

γ . Parameter-Efficient Fine-Tuning: Adapters and LoRA

As models grow in scale, updating all parameters during fine-tuning becomes computationally prohibitive. To address this, **Parameter-Efficient Fine-Tuning** (PEFT) [Xu et al. \(2023\)](#) techniques have been developed to adapt large pre-trained models using only a small fraction of trainable weights.

1. Adapters

Adapter-based tuning [Houlsby et al. \(2019\)](#) involves inserting small bottleneck layers between existing modules of a Transformer block.

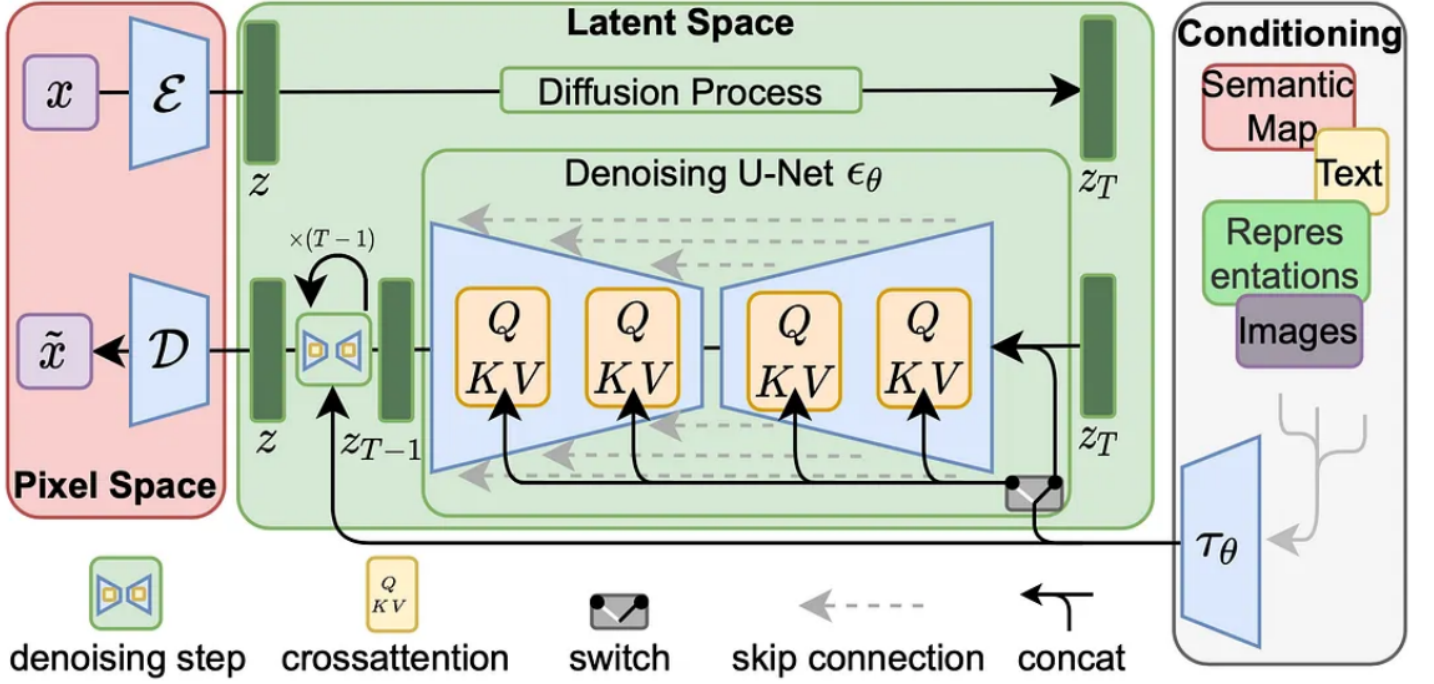


Figure 9: Architecture of the Latent Diffusion Model. The denoising U-Net incorporates cross-attention mechanisms to integrate semantic conditioning (τ_θ), allowing for guided image synthesis in the compressed latent space.

- **AdapterH**: The original design proposed by [Houlsby et al. \(2019\)](#), which inserts layers after both the multi-head attention and the MLP modules.
- **AdapterL** ([Houlsby et al., 2019](#)) : A more streamlined version where the adapter is applied only after the MLP module to reduce latency.
- **AdapterDrop** ([Rücklé et al., 2020](#)): An efficiency-focused approach that selectively removes adapter layers from the lower blocks of the network to accelerate inference.

2. Low-Rank Adaptation (LoRA)

While adapters add extra layers that can increase inference latency, Low-Rank Adaptation (LoRA) ([Hu et al., 2021](#)) offers a more seamless integration. Instead of adding new layers in sequence, LoRA injects trainable low-rank decomposition matrices directly into the pre-trained weights, typically within the self-attention sub-layers.

For a pre-trained weight matrix $W_0 \in \mathbb{R}^{d \times k}$, LoRA represents its update ΔW through the product of two low-rank matrices $A \in \mathbb{R}^{d \times r}$ and $B \in \mathbb{R}^{r \times k}$, where the rank $r \ll \min(d, k)$. As illustrated in Figure 10 :

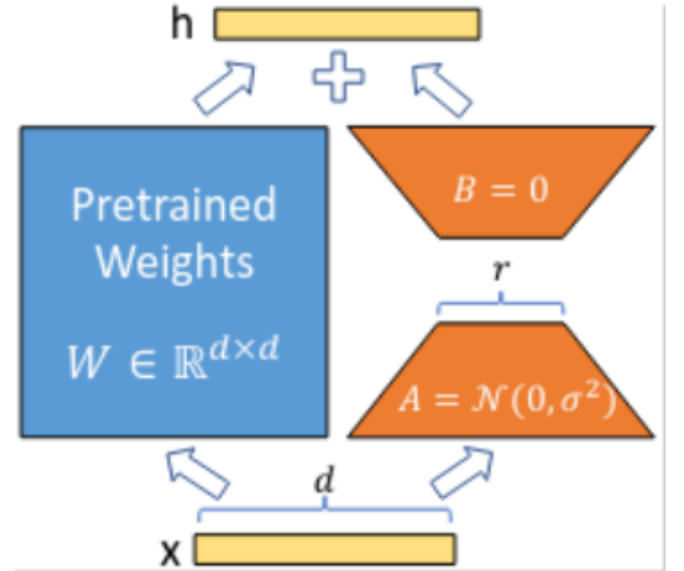


Figure 10: The LoRA architecture : W_0 remains frozen, and only A and B are updated [Hu et al. \(2021\)](#)

$$W = W_0 + \Delta W = W_0 + BA \quad (8)$$

During training, W_0 remains frozen, and only A and B are updated. This approach is particularly effective for

δ . Reward-Based Learning and Judge Models

Beyond standard supervised training on fixed labels, aligning generative models ([Nareklshvili et al., 2024](#))

with human-centric criteria like aesthetic quality, stylistic consistency, or safety often requires **Preference Fine-Tuning** [Dang et al. \(2025\)](#). This paradigm shift moves away from minimizing a reconstruction loss toward maximizing a reward signal that reflects specific desired attributes.

1. Reward Models as Proxies for Human Judgment

Direct human evaluation is the "gold standard" for assessing visual style, but it is slow, expensive, and non-scalable for iterative training loops. To address this, **Reward Models** [Zhong et al. \(2025\)](#) (also known as **Judge Models**) are trained to act as automated proxies for human experts. These models learn a mapping from the generated output space to a scalar reward score, effectively quantifying how well an image adheres to a specific convention, such as the **ARASAAC** standard.

2. Optimization via Feedback

In modern pipelines, these scores guide the optimization of the generator-often through **Reinforcement Learning from Human Feedback** (RLHF) [Lambert \(2025\)](#), or Direct Preference Optimization (DPO) [Rafailov et al. \(2023\)](#).

In the context of Parameter-Efficient Fine-Tuning [Xu et al. \(2023\)](#), the reward signal is used to weight the gradient updates of the **LoRA** matrices. This ensures that the model does not just learn to generate any image, but specifically optimizes for the stylistic and structural requirements identified by the Judge, such as edge sharpness and visual simplicity.

3 THE SEMA PIPELINE

The core objective of our framework is to transform a standard generative model ([Nareklishvili et al., 2024](#)) into a specialized engine capable of producing pictograms that are both semantically accurate and stylistically compliant with the ARASAAC convention.

To achieve this, we developed a multi-stage pipeline -the SEMA Pipeline- which bridges the gap between natural language and structured iconography.

The primary challenge lies in the fact that general-purpose diffusion models naturally gravitate towards visual complexity and realistic textures.

By implementing the four-stage process illustrated in [Figure 11](#), we move from raw data to a "**style-aware**" **system**. This structure ensures that every generated symbol is not only a visual representation of a concept but also

a functional tool for **Augmentative and Alternative Communication** (AAC).

The pipeline is organized into the following key phases:

1. **Data Alignment:** We first build a high-quality foundation by pairing raw ARASAAC icons with precise, synthetically generated captions to ensure the model understands the specific visual-semantic relationship of pictograms.
2. **Reward Model (The Judge):** Simultaneously, we train a dedicated MLP-based "Judge." This model learns to quantify the "ARASAAC-ness" of an image by analyzing geometric and perceptual features such as edge sharpness and visual entropy.
3. **Guided Fine-Tuning:** We use Low-Rank Adaptation (LoRA) to update the Stable Diffusion model ([Oorloff et al., 2025](#)). Crucially, the fine-tuning process is supervised by the Judge, which provides a reward signal to steer the model toward the desired minimalist style.
4. **Inference and Validation:** In the final stage, images are generated and passed through a final filtering (validation system) to ensure high-fidelity, standardized pictograms reach the end-user.

Phase 1: Automated Image-to-Text Labeling

A major challenge in this research was the lack of high-quality descriptive captions for existing ARASAAC datasets. Standard pictogram libraries are often indexed by single-word keywords, which are insufficient for the cross-attention mechanisms of Latent Diffusion Models.

The SEMA-Labeler Architecture

The initial development phase focused on creating a "zero-shot" semantic bridge, which we named the **SEMA-Labeler**. The goal was to generate captions that were not just labels, but architectural descriptions of the pictograms.

As shown in [Figure 12](#), the pipeline was built on three technical pillars:

- **Multimodal Extraction:** We utilized the **CLIP** (ViT-L/14) ([Radford et al., 2021](#)) encoder to map the visual structure of ARASAAC icons into a high-dimensional embedding space.
- **Lexical Enrichment:** Using a custom script, we cross-referenced ARASAAC tags with linguistic dictionaries (wn-3.1) to provide context to the LLM.

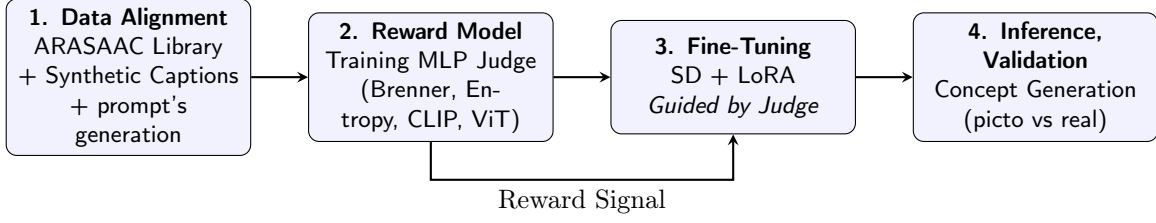


Figure 11: The SEMA end-to-end pipeline: From raw symbol alignment to validated iconographic generation.

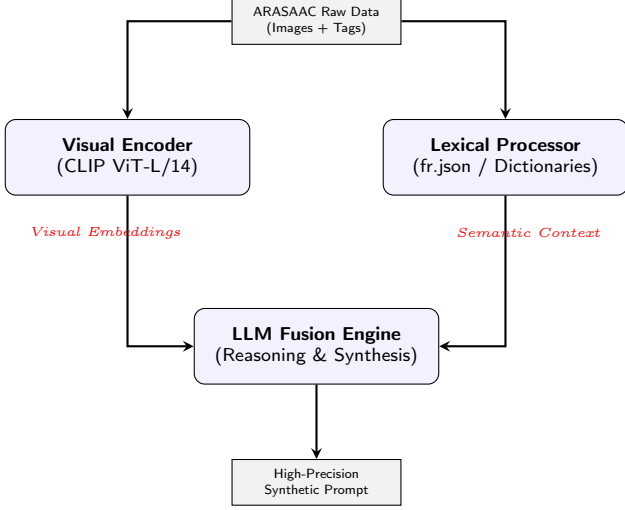


Figure 12: The proposed SEMA-Labeling Architecture. This design aimed to fuse high-dimensional visual embeddings with cleaned lexical metadata to generate descriptive training prompts.

- **Logic Fusion:** The system aimed to feed these visual embeddings and cleaned tags into an LLM to "reason" about the symbol's meaning (e.g., distinguishing between "eating" as an action and "food" as a category).

While the architecture was conceptually sound, its implementation faced critical hardware and software hurdles:

- **VRAM and Memory Constraints:** Loading the high-rank CLIP encoders alongside the LLM inference engine lead to persistent *Out-of-Memory* (OOM) errors on the available hardware.
- **Dependency Conflicts:** Incompatibilities between the specific library versions required for the vision-encoders and the LLM wrappers caused significant integration delays.
- **Time Efficiency:** Processing 12,000+ images with this deep fusion approach was estimated to exceed the remaining phase timeline.

Implementation via Adopted Pipeline

To ensure the continuity of the full SEMA pipeline and proceed to the fine-tuning stage, we integrated the labeling work developed by Favre (2024). This system allows us to maintain the project's momentum by utilizing a two-step distillation process:

- **Visual Description:** Using **Qwen-VL-8B** as the "Eyes" to analyze the physical shapes and objects within each ARASAAC icon.
- **Semantic Refinement:** Using **GPT-OSS-120B** as the "Brain" to convert these raw descriptions into standardized, clean training prompts (see Figure 13).

By adopting this solution for the dataset alignment, we successfully paired the visual ARASAAC style with standardized textual prompts, creating the essential ground truth required for the subsequent LoRA fine-tuning.

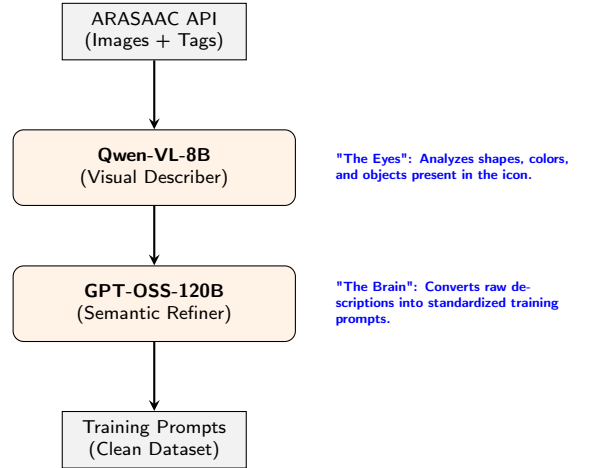


Figure 13: The two-step distillation pipeline. By separating visual description from semantic refinement, we ensure high-quality prompts for the Diffusion model.

The results of this automated Image-to-Text alignment, as exemplified by the samples in Figure 14, provide the high-quality ground truth necessary for both the training of the Reward Model and the subsequent stylistic adaptation of Stable Diffusion.

Pictogram Generated Synthetic Prompt Favre (2024)'s Pipeline



["bee"] Hairy-bodied black-and-yellow striped insect with translucent wings, antennae, and pollinating work.



["borage"] Annual herb with hairy leaves, blue flowers, used fresh in salads and culinary dishes.



["frogman", "diver", "scuba diver"] Underwater diver equipped with scuba gear, mask, fins, and breathing apparatus.



["cannellonis"] Broad tubular pasta baked with meat or cheese filling, topped with sauce and cheese.

Figure 14: Examples of paired data obtained after the Distillation Phase. These detailed prompts allow the Diffusion model to learn the specific stylistic features of the ARASAAC standard.

Phase 2: Stylistic Validation and the MLP Judge

The core of our stylistic control lies in a dedicated Reward Model (Zhong et al., 2025), or "Judge," designed to act as an automated arbiter of ARASAAC compliance. This model transforms subjective visual style into a quantifiable, differentiable signal.

1. The Stylistic Feature Vector

a. Geometric Integrity: Brenner Gradient (v_{br})

The Brenner gradient F.brenner and Brock. (1976) is a focus measure used to quantify the sharpness of edges. Pictograms are characterized by high-frequency transitions at the boundaries of shapes (bold black outlines), unlike the soft gradients of natural photographs. The score is computed by summing the squared differences between pixels at a distance of two:

$$f_{Brenner} = \sum_{x,y} [I(x+2,y) - I(x,y)]^2 \quad (9)$$

A high v_{br} value ensures the visual "readability" required for augmentative communication.

b. Visual Simplification: Shannon Entropy (v_{ent})

To prevent the generation of unnecessary textures, we employ Shannon Entropy (Vajapeyam, 2014) as a proxy for visual complexity. Given the probability distribution P of gray levels, the entropy is:

$$H = - \sum_{i=0}^{255} P_i \log_2(P_i + \epsilon) \quad (10)$$

Standardized pictograms possess low entropy due to their extensive use of uniform color fields. Minimizing H forces the generator to adhere to the minimalist ARASAAC convention.

c. Semantic Grounding: CLIP Similarity (v_{cl})

We utilize the CLIP Radford et al. (2021) framework to ensure that stylistic abstraction does not supersede semantic meaning.

We calculate the cosine similarity between the visual embedding E_{img} and the textual embedding E_{txt} . While the raw model output (S_{CLIP}) can exceed 1.0 due to the learned temperature parameter e^t , we normalize this value back to the $[-1, 1]$ hyperspace for metric consistency:

$$\text{sim}(E_{img}, E_{txt}) = \frac{E_{img} \cdot E_{txt}}{|E_{img}| |E_{txt}|} \quad (11)$$

$$S_{CLIP} = \text{sim}(E_{img}, E_{txt}) \cdot e^t \quad (12)$$

In our implementation, the feature vector V uses the normalized $\text{sim}(E_{img}, E_{txt})$ as v_{cl} . This ensures that the semantic alignment score remains a robust focus for the Z-score normalization layer, preventing the large magnitude of the logits from overwhelming the geometric metrics.

d. Object Recognition: MobileViT Confidence (v_{sn})

We chose MobileViT Mehta and Rastegari (2021) -as our object recognition model- because it provides strong image-recognition performance with a compact, efficient backbone (convolutional inductive bias + lightweight

self-attention), making it well-suited for evaluating low-resolution, icon-like inputs at scale without excessive computational cost. We extract the *maximum softmax probability* (top-1 confidence):

$$v_{sn} = \max(\text{Softmax}(\mathbf{z})) \quad (13)$$

High confidence implies the iconographic representation is unambiguous and universally decodable.

2. Non-Linear Arbitrated Scoring

The *-Multi Layer Perceptron- MLP Judge* serves as the final decision-making layer of our framework, acting as a **multi-criteria** discriminator that synthesizes visual and semantic data into a single quality signal.

By ingesting a feature vector composed of the CLIP Score (Radford et al., 2021), Shannon Entropy (Vajapeyam, 2014), the Brenner Gradient (F.brenner and Brock., 1976), and the MobileViT Confidence (Mehta and Rastegari, 2021), the model evaluates the degree of alignment between the generated output and the strict ARASAAC iconographic standards.

This 4-dimensional input is processed through a sequential architecture of dense layers (16 and 8 neurons) using **ReLU activations** to capture non-linear correlations between sharpness and stylistic purity.

The process culminates in a Sigmoid activation function, which maps the internal representations to a unified reward score $S \in [0, 1]$, providing a continuous feedback loop essential for the stable fine-tuning of the LoRA weights.

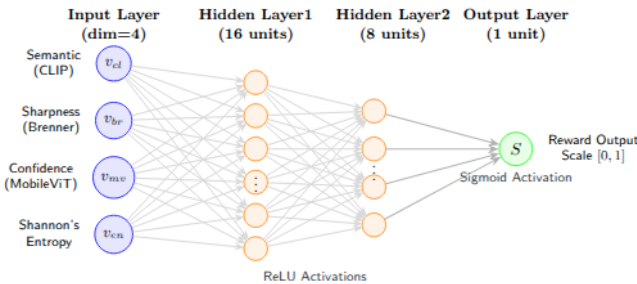


Figure 15: The Architecture of our MLP

3. Data Augmentation via α -Interpolation

A significant challenge was the binary nature of the raw training data (Real images vs. Official Pictograms). To create a continuous decision boundary, we implemented an **interpolation strategy**. We mixed the feature vectors using a coefficient $\alpha \in [0, 1]$:

$$V_{train} = \alpha V_{picto} + (1 - \alpha) V_{real} \quad (14)$$

This allows the MLP to learn the "slope" of stylistic transition, providing a scalar score that reflects the *degree* of compliance rather than a simple binary classification.

4. Input Normalization and Stability

A critical preprocessing step before feeding the feature vector $\mathbf{V} = [v_{br}, v_{ent}, v_{cl}, v_{sn}]$ into the MLP Judge is the application of a Z-score normalization. Given the disparate numerical ranges of the extracted metrics, each component v_i is transformed as follows:

$$\hat{v}_i = \frac{v_i - \mu_i}{\sigma_i} \quad (15)$$

Where μ_i and σ_i represent the mean and standard deviation of the i -th metric, specifically computed across our entire balanced training dataset (2,400 samples).

The characteristic statistics for each metric are summarized in Table 3:

These statistics (μ, σ) are frozen after the initial dataset analysis. By keeping these parameters constant within the Judge's architecture, we ensure a stable and consistent reward signal during the iterative LoRA fine-tuning process, preventing distribution shifts that could degrade stylistic convergence.

Table 3: Normalization Statistics for the MLP Judge

Metric	Brenner	Entropy	CLIP	mobile ViT
Mean (μ)	19.5338	4.6595	22.5709	0.4282
Std (σ)	12.7541	2.1522	3.9989	0.2393

Phase3 : Fine-Tuning Integration & LoRA Reward Weighting

The final stage of the **SEMA** (Multimodal Framework for Semantic Alignment and Controlled Pictogram Generation) framework involves the integration of the MLP Judge's output into the Diffusion training loop.

1. Reward-Weighted Fine-Tuning:

We propose a Reward-Weighted Fine-Tuning (RWFT) (Fan1 et al., 2025) approach, where the MLP score $S \in [0, 1]$ dynamically modulates the **standard Latent Diffusion Loss**.

This loss is defined by the Mean Squared Error (MSE) between the added noise ϵ and the noise predicted by the model (?):

$$\mathcal{L}_{ld} = \mathbb{E}_{z, \epsilon, t, y} [|\epsilon - \epsilon_\theta(z_t, t, y)|^2] \quad (16)$$

To find the optimal convergence balance, three distinct strategies for the total objective function $\mathcal{L}_{total}$ were evaluated:

1. **Baseline (Standard LoRA):** The model is trained without reward modulation to establish a performance floor:

$$\mathcal{L}_{total} = \mathcal{L}_{ld} \quad (17)$$

2. **Linear Inverse Error Weighting:** This is the primary proposed strategy, where the loss is scaled by the stylistic error $(1 - S)$.

$$\mathcal{L}_{total} = \lambda \cdot (1 - S) \cdot \mathcal{L}_{ldm} \quad (18)$$

The hyperparameter λ acts as the "gain" of the stylistic feedback loop, determining how aggressively the MLP Judge influences the model's weights:

- **Conservative Range** ($\lambda \in [1, 5]$): Ensures semantic stability. It allows the model to retain complex knowledge from the base Stable Diffusion while slowly adopting the ARASAAC style. This is ideal for maintaining the "meaning" of difficult concepts.
- **Aggressive Range** ($\lambda \in [5, 10]$): Forces stylistic compliance. These values accelerate the transition to "Flat Design" and thick outlines. However, they carry a risk of Mode Collapse, where the model may prioritize simple shapes over semantic accuracy to satisfy the Judge.

3. **Smooth Hyperbolic Gradient Scaling** A hyperbolic penalty strategy designed to aggressively override the "realism bias" of the base model:

$$\mathcal{L}_{total} = \frac{1}{S + \epsilon_{stability}} \cdot \mathcal{L}_{ld} \quad (19)$$

Where $\epsilon_{stability} = 10^{-6}$ is a stability constant used to prevent gradient explosion and ensure numerical robustness.

2. Numerical Stability and Gradient Explosion

While the **Smoothed Reciprocal Weighting** is theoretically powerful, it proved highly unstable for "Cold Starts". At the beginning of training (Epoch 1), the generated images typically yield a score of $S \approx 0$.

- **The Phenomenon:** With S near zero, the loss multiplier reached $1/\epsilon$ (e.g, 10^5), triggering an immediate **Gradient Explosion**.
- **The Result:** This massive update pushed the LoRA weights into extreme ranges, resulting in **Nan (Not a Number)** loss values and a total collapse of the model's latent space.
- For initial training phases, **Linear Inverse Error Weighting** ($1 - S$) is preferred, as it caps the maximum gradient multiplier at 1.0, ensuring numerical robustness.

3. Evaluation Metrics : KID & Judge

To assess the impact of our Reward-Weighted Fine-Tuning (RWFT), we utilize two complementary metrics that balance individual sample quality with global stylistic distribution.

- **Judge Score** ($S \in [0, 1]$): An internal metric representing the MLP Judge's confidence in stylistic compliance. The objective is to maximize this score ($S \rightarrow 1$), indicating optimal alignment with sharpness, complexity, and ARASAAC-style constraints.
- **Kernel Inception Distance (KID):** Bińkowski et al. (2018) A distribution-level metric that compares the features of a batch of authentic ARASAAC icons (P_r) against a batch of generated icons (P_g).

Mathematically, the KID is defined as the squared **Maximum Mean Discrepancy (MMD)** between these two batches:

$$KID(P_r, P_g) = |\mu(P_r) - \mu(P_g)|_{\mathcal{H}}^2 \quad (20)$$

In this formulation, μ represents the mean embedding of the images in a high-dimensional feature space (Hilbert space $\mathcal{H}$). A KID of 0 implies that the model’s output batch is statistically indistinguishable from the authentic dataset batch.

4. Base Architectures for Fine-Tuning

We evaluated two distinct latent diffusion checkpoints to determine the impact of the model’s "prio" on stylistic convergence:

1. **Stable Diffusion v1.5 (RunwayML)**: A standard foundation model (Huyen, 2025) with a massive semantic latent space but a strong photographic bias. It served as the primary benchmark for measuring the MLP Judge’s ability to "break" realism in favor of iconography.
2. **AnyLoRA (Lykon)**: A checkpoint specifically optimized for stylistic flexibility and 2D illustration. This model was selected due to its neutralized weights, which facilitate faster convergence toward the ARASAAC flat-design distribution with lower reward pressure.

Phase4: Fine-Tuning Results

1. Results and Analysis

As illustrated in Figure 17, the quantitative analysis reveals a significant **temporal decoupling** between stylistic compliance and distributional fidelity.

While the **AnyLoRA** ($\lambda(1 - S)$) configuration appears to be the superior candidate, its "Best" values are achieved at different stages: the optimal **KID (0.059)** is reached at Epoch 1, whereas the **Best Judge Score (0.981)** is only attained at Epoch 10 (figure 16).

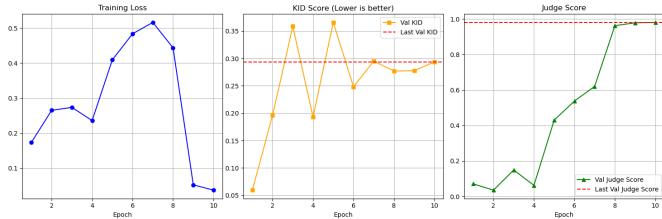


Figure 16: Training metrics for the AnyLoRA configuration with $1 - S$ reward weighting. From left to right: (a) Training Loss, (b) Kernel Inception Distance (KID), and (c) MLP Judge Score. Note the optimal KID at Epoch 1 vs. the maximum Judge Score at Epoch 10, illustrating the stylistic-semantic trade-off.

The temporary increase in training loss observed between Epoch 4 and 7 reflects the tension between the diffusion objective and the stylistic constraint ($1 - S$). As the model navigates away from its pre-trained realistic priors, the reward-weighting mechanism penalizes non-compliant intermediate representations, leading to a transient loss elevation.

The subsequent sharp decline after Epoch 7 coincides with the saturation of the Judge Score ($S \rightarrow 1$), effectively nullifying the loss magnitude as the model converges toward a high-reward, yet potentially over-specialized, stylistic regime.

This implies that while the model masters the ARASAAC geometric primitives late in training, it maintains the closest statistical resemblance to the original dataset early on.

Furthermore, the results highlight the severe **numerical instability** of the reciprocal weighting strategy ($1/\epsilon + S$). Despite achieving a near-perfect Judge Score (0.991) at Epoch 17, this configuration triggered a **Gradient Explosion** almost immediately, resulting in a **NaN (Not a Number) Loss**.

Crucially, even after 17 epochs of training, the loss remains non-finite, indicating that the model’s latent space has effectively collapsed. While the MLP Judge is "fooled" by the high-contrast artifacts produced by the exploding gradients, the catastrophic KID score (0.778) confirms that the semantic integrity of the images has been destroyed, rendering this strategy unviable for stable pictogram generation.

2. Inference and Results Analysis

This section evaluates the qualitative performance of the fine-tuned models. We compare AnyLoRA model against various Stable Diffusion (SD) configurations and the reciprocal loss function ($1/S$).

α . Comparative Analysis

The table 18 illustrates the trade-off between stylistic enforcement and semantic integrity.

β . Success Cases

The figure 19 represent instances where the model successfully transitioned from the photographic domain to a structured, illustrative domain suitable for assistive communication, despite the remaining complexity.

These samples represent the current upper bound of the model’s performance. **While they do not yet achieve the extreme minimalist 'flat-design' of ARASAAC**, they demonstrate a successful shift from photorealism to a structured illustrative domain.

The preservation of complex semantic features (e.g., the

Model Name	Rank	λ	KID Samples.	LR	Judge $\uparrow$ (epoch)	KID $\downarrow$ (epoch)	Best Loss
AnyLoRA Baseline	8	0	300	1e-5	0.043 (2)	0.062 (7)	0.03
SD v1.5 Baseline	16	0	200	1e-5	0.075 (4)	0.062 (2)	0.04
SD v1.5 + $\lambda^*(1-S)$	8	3.0	200	1e-4	0.937 (5)	0.071 (7)	0.062
AnyLoRA + $\lambda^*(1-S)$	8	5.0	200	1e-4	0.981 (10)	0.059 (1)	0.04
SD v1.5 + $\lambda^*(1/\epsilon+S)$	16	2.0	200	1e-5	0.991 (17)	0.778 (6)	Nan

Figure 17: Training Dynamics of Reward-Weighted Fine-Tuning

Notion & Target Prompt	GT (ARASAAC)	Model Configuration	Generation	Critical Analysis
Medal, Gold medal <i>"Award recognizing championship victory or commemorating an event, often gold with laurel."</i>		AnyLoRA (1-S) best judge		Partial Success: The model shifts toward a drawing style, but fails to reach ARASAAC's minimalism. It produces a complex illustration with many colors and details, whereas the GT uses flat areas and very few lines.
		SD with (1-S) best judge		Structural Failure: Compared to the clear silhouettes of the GT, this model only generates a circular gold texture. The human figure and the concept of "winning" present in ARASAAC are completely lost.
		SD with (1-S) best KID		Style Mismatch: The output is a realistic 3D object with shadows and depth. It ignores the 2D flat-design constraint seen in the ARASAAC examples, remaining too close to a real-world photograph.
		SD (1/S)		Total Divergence: The generation results in unreadable digital noise. Unlike the functional and clear GT icons, this output provides no visual information and represents a complete training collapse.

Figure 18: Comparative synthesis of stylistic enforcement across different architectures and loss functions. The AnyLoRA (1-S) configuration demonstrates the best trade-off between illustrative domain transfer and semantic retention, whereas the reciprocal (1/S) loss leads to catastrophic structural failure.

internal structure of the brain or the mechanical components of the watch) proves that our SEMA approach maintains high data fidelity while suppressing base model skeuomorphic priors

γ . Failure Modes

While the fine-tuning process demonstrated convergence on key metrics (Judge Reward and KID), visual inspection of inference outtakes reveals persistent edge-case failures.

The following analysis categorizes the primary obstacles observed during the inference phase, referencing the 9-sample diagnostic grid (figure 20).

These failures highlight the conflict between base model priors and the target minimalist iconography.

a. Domain Entrapment (Photographic Bias)

Despite the fine-tuning, the model often remains tethered to its original photographic training.

- **Image [2,1] - Retail Store:** The model generates a complex 3D environment with realistic lighting and depth, completely ignoring the 2D flat-design requirement.
- **Image [2,3] - Portrait:** The output is a high-fidelity realistic character, demonstrating the "Base Model Prior" dominance over the stylistic instructions.

b. Semantic Drift and Reward Hacking

The model sometimes "cheats" the Reward Function by producing sharp textures that satisfy the Judge but lose the intended meaning.

- **Image [1,2] - Hand Interaction:** Structural collapse of the hand geometry. The semantic connection between the person and the object is lost in favor of unreadable textures.
- **Image [3,1] - Reward Hacking:** The model prioritizes high-contrast metallic reflections (gold texture) over the actual human silhouette or action described in the prompt.

c. Mathematical Instability(NaN Collapse)

The use of reciprocal loss functions leads to numerical divergence in the latent space.

Image [2,2] - Digital Noise: A complete failure of the generation process. The image is replaced by saturated

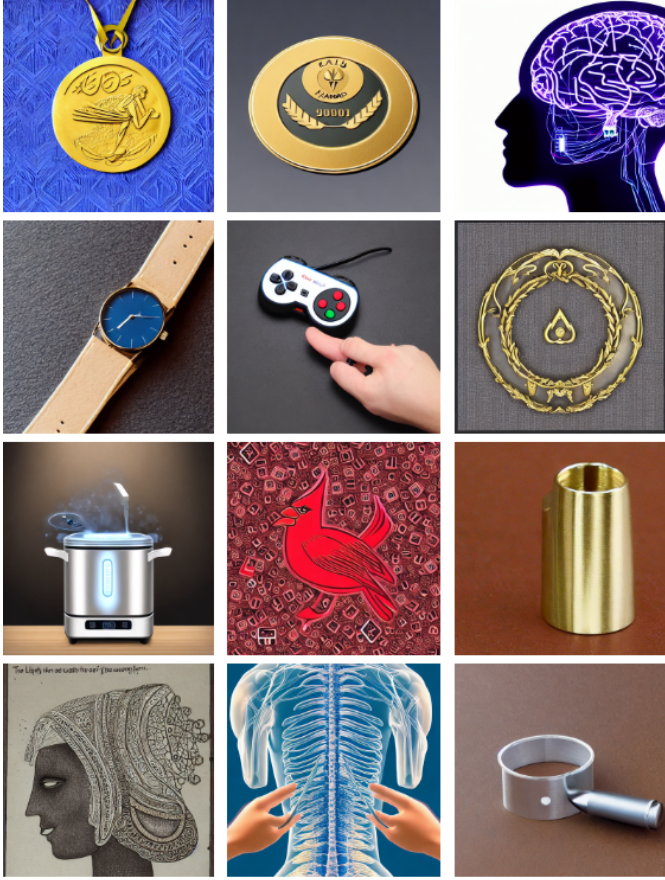


Figure 19: Qualitative gallery of successful domain transfers. These samples exhibit clean outlines and coherent geometry, representing a significant shift from photographic priors toward a structured, illustrative style compatible with AAC requirements.

pixels and high-frequency artifacts, a direct result of gradient explosion during training.

d. The Illustration-Icon Gap

Even when a non-photographic style is achieved, the model struggles with extreme simplification.

Image (1,3) - Gamepad: While successful as a 2D vector drawing, it remains an *illustration* with too many details. It fails the "Functional Icon" test which requires the minimalist abstraction seen in ARASAAC.

Future Work and Extensions

While the SEMA framework establishes a robust baseline for controlled pictogram synthesis, several avenues for extension remain to be explored:

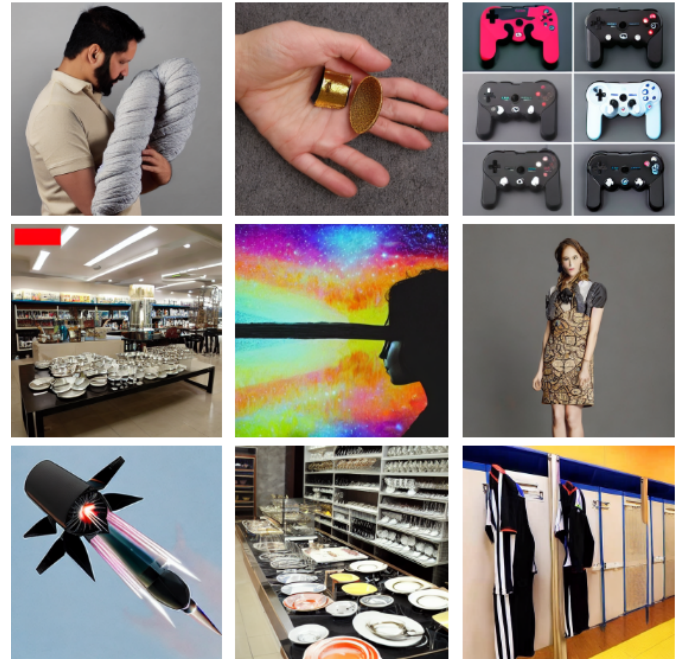


Figure 20: Visual categorization of persistent failure modes. This 3x3 diagnostic grid highlights (a) semantic drift due to reward hacking, (b) stylistic stagnation where the model remains trapped in 3D realism, and (c) total latent space collapse (digital noise) resulting from gradient instability.

1. Training from Scratch: The Hybrid 50/50 Approach

A significant limitation of fine-tuning is the "memory" of the base model's realistic priors. We propose training a **Small-Scale Diffusion Model from scratch** using a balanced hybrid dataset: 50% authentic ARASAAC pictograms (approx. 13,000 samples) and 50% curated real-world images. By starting from a neutral initialization, the model could learn a shared latent space where geometric primitives (lines, flat colors) are as fundamental as photographic features, potentially easing the transition to extreme minimalism.

2. Architectural Simplification: Beyond Diffusion

Given that pictograms are structurally simpler than natural scenes, the high-dimensional complexity of a full Diffusion Pipeline may be redundant. We propose two alternative architectural paths:

- **Standalone VAE Bottleneck:** Utilizing a Variational Autoencoder (VAE) alone to learn a highly compressed latent space dedicated to 2D shapes. Since pictograms lack complex textures, a VAE could act as a generative manifold without the need for an iterative denoising process.

- **Pure U-Net Mapping:** Testing a direct U-Net translation architecture (similar to Pix2Pix (Isola et al., 2018)) to map semantic embeddings directly to pixel grids. This would significantly reduce inference latency and prevent the stylistic "drifts" often observed in multi-step diffusion.

3. Refined Prompt Structuring

Current results suggest that standard descriptive prompts are insufficient for iconographic control. A "Structure-Aware" labeling strategy could be implemented, where prompts are formatted as structural recipes (e.g., "*Shape: Circular; Outline: Bold; Concept: Action_Eat*"). This would allow the Cross-Attention mechanism to treat style and semantics as independent orthogonal vectors.

4. User-in-the-Loop Validation

Integrating **Human-in-the-loop (HITL)** feedback into the MLP Judge would allow the reward signal to align with actual AAC user comprehension. Training the Judge on real-world eye-tracking data or recognition speed tests would ensure that "quality" is defined by accessibility rather than just geometric metrics.

5. Semantic-Preserving CLIP Penalty

To mitigate "Reward Hacking" (where the model creates abstract textures to satisfy edge-sharpness metrics), we propose integrating a **Directional CLIP Loss**. This would penalize the model if the embedding of the generated image deviates beyond a specific threshold from the Ground Truth ARASAAC embedding, ensuring that stylistic shift never occurs at the expense of meaning.

6. Entity-Driven Conditioning via NER Filtering

To reduce the semantic noise inherent in natural language prompts, we propose a conditioning mechanism guided by our **NER benchmark**. Instead of feeding the raw prompt into the Diffusion cross-attention layers, the text is first processed by a **NER** model to extract key semantic entities (e.g., *Person, Object, Location, action, city*).

By concatenating these structured entity embeddings with the global CLIP representation, the model can "focus" its generative attention on the essential components of the pictogram.

This prevents the generation of irrelevant background details and ensures that the U-Net prioritizes the "archetype" identified by the NER, leading to a more functional and minimalist visual output aligned with ARASAAC standards.

CONCLUSION

This framework demonstrates that combining deep semantic understanding (via Pantagruel and NER) with rigorous statistical validation (via the MLP Judge) enables the production of accessible, standardized pictograms at scale. By leveraging French-specific language models, the SEMA pipeline ensures that linguistic nuances are preserved during the transition from text to iconography. Ultimately, this approach bridges the gap between generative AI and the strict accessibility requirements of Augmentative and Alternative Communication (AAC).

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